Geophys. J. R. astr. Soc. (1985) **83**, 721–729

Palaeomagnetic study of an allochthonous terraine: the Scisti Silicei Formation, Lagonegro Basin, southern Italy

A. Incoronato* and D. H. Tarling *Department of Geophysics and Planetary Physics, The University, Newcastle upon Tyne NE1 7RU*

G. Nardi *Dipartimento Scienze della Terra, Università degli Studi di Napoli, largo S. Marcellino, 10, 80138 Naples, Italy*

Accepted 1985 April 17. Received 1985 March 25; in original form 1985 October 23

Summary. The Jurassic Scisti Silicei Formation forms part of the Lagonegro superimposed tectonic units I and II that are thought to represent the axial and internal margins of the Mesozoic Lagonegro Basin, prior to nappe formation. Sampling was carried out in the lower (Lagonegro) and upper (Pignola) nappes in two differently oriented anticlines. Single and multi-component magnetizations are present. Isothermal remanence acquisition rates show that magnetite and haematite are present which, in most Lagonegro specimens, show the same direction of magnetization. Comparison of the palaeomagnetic directions with those from Jurassic rocks on the stable African craton indicates a 147° anticlockwise rotation of the lower nappe which is similar to 139° previously reported for the upper nappe at Vietri di Potenza. The same comparisons show a 44° clockwise rotation of the upper nappe at Pignola. These results suggest that the doubled nappe structures, sampled some 50 km apart, resulted from their emplacement by translation with little rotation prior to the opening of the Tyrrhenian Sea and that it was the opening of this Sea that caused the predominantly anticlockwise rotation. This work therefore indicates the way in which palaeomagnetic analyses can be used, even within complex allochthonous areas, as an aid to deciphering their tectonic evolution.

Key words: Italy, Jurassic, Mediterranean, palaeomagnetism, tectonics

Introduction, geology and sampling

Most previous palaeomagnetic studies in Italy have been concerned with either magneto-stratigraphy (Lowrie *et al.* 1980; Lowrie & Channell 1984) or definition of the extent and development of the (geologically African) promontory of 'Adria' (Argand 1924; Channell & Tarling 1975; Channell *et al.* 1979; D'Argenio, Horvath & Channell 1980) as this tectonic

* Now at: Dipartimento Scienze della Terra, Università degli Studi di Napoli, largo S. Marcellino, 10, 80138 Naples, Italy.

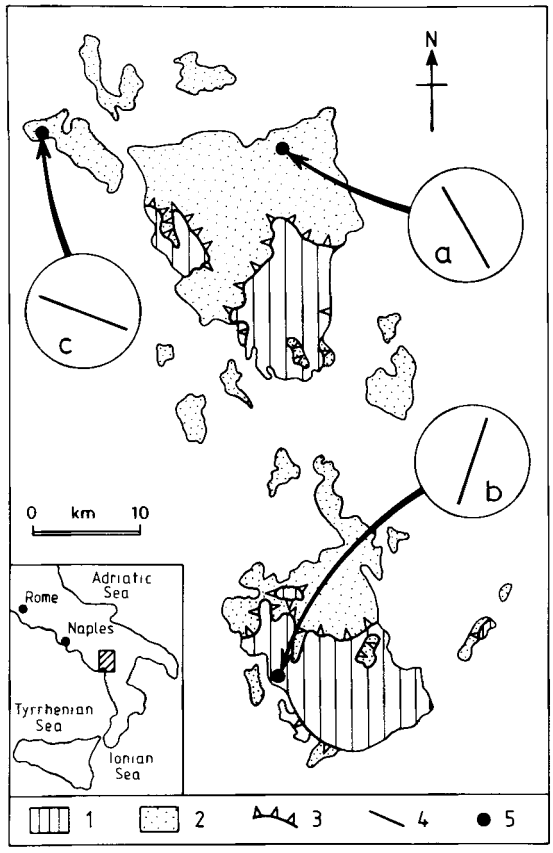


Figure 1. Simplified geological map of the Lagonegro Basin, Lucania.
(1) Lagonegro Unit I (lower nappe).
(2) Lagonegro Unit II (upper nappe).
(3) Thrust.
(4) Present-day local orientation of anticlinal axes.
(5) Site locations (a) Pignola, (b) Lagonegro, and (c) Vietri di Potenza.

unit is fundamental to an understanding of the evolution of the Alps. In this paper, palaeomagnetic techniques have mainly been used to investigate the structural evolution within a complex allochthonous area which has been subjected to intensive geological study within a continuing review of the overall evolution of the southern Apennine chain (D'Argenio, Pescatore & Scandone 1975; Catalano & D'Argenio 1978). These results are also of interest as they demonstrate that a variety of highly siliceous sediments can be suitable for palaeomagnetic study.

The rocks of the Lagonegro Basin outcrop mainly in the Lucania region (Fig. 1) and are part of the series of décollement nappes which form the Southern Apennine Chain. The Basin developed since mid-Triassic times as a result of rifting between Africa and Europe which also marked the first outline of 'Adria' (Scandone 1975; D'Argenio 1976). The area then formed part of a carbonate platform/siliceous basin system which paralleled the African palaeomargin and appears to be analogous to the present-day Bahamas region (D'Argenio 1970; Bernoulli 1972; Bernoulli & Jenkins 1974). During the Mesozoic, different tensional and transcurrent tectonic phases affected the area until, in Lower-Middle Miocene time, the western edge of Adria was affected by strong compression resulting in the formation of a

1365246xb, 1985, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1985.tb04334.x by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

huge stack of nappes which now form the Southern Apennines. Within this complex structure, different stress fields, with consequently different fold axes orientations, occurred at different times. In the Aquitanian–Langhian, the axes were oriented E–W and showed a northern vergence, but the axes paralleled the western margin in the Tortonian and the degree of folding became increasingly intense. One possible model for their overall structure is that of an initial anticlockwise rotation of the nappes, related to the opening of the Tyrrhenian Sea during the Tortonian–Messinian (Di Nocera, Ortolani & Torre 1975; Catalano *et al.* 1976; Ortolani 1978), with the western Basin margin being thrust over its eastern parts in the Aquitanian–Langhian (Scandone 1972). This doubled sequence was then moved even further eastwards, causing the axes of the folding to become concave towards the Tyrrhenian Sea.

Such a structural model for the evolution of these nappe structures depends strongly on the occurrence of rotations during their transport and hence should be susceptible to testing by palaeomagnetic methods (Tarling 1983). This study reports results from the Jurassic Scisti Silicei Formation which outcrops in two superimposed nappes on the southern and central parts of the overall arcuate structure (Fig. 1). This formation is also coeval with the rocks sampled in the upper nappe by Channell (1975) in the north-eastern part of the arc. Twenty-five sites (six samples per site) were sampled in red and green jaspers, siliceous claystones and graded calcareous breccias and marls in the basal part of lower nappe at Pignola. These samples were highly siliceous and fractured, preventing drilling in the field, with the exception of four sites. Most samples were collected as small hand samples which were shaped into 2.1 cm sided cubes and spheres (Incoronato 1982). A further 24 sites, comprising silicified marls and clays, with polychromous jaspers, were drilled from the basal part of the upper nappe at Lagonegro.

Measurement and analysis

All specimens were weakly magnetized and six sites from Pignola and three from Lagonegro were unmeasurable even with a cryogenic magnetmeter. The remaining Pignola samples had

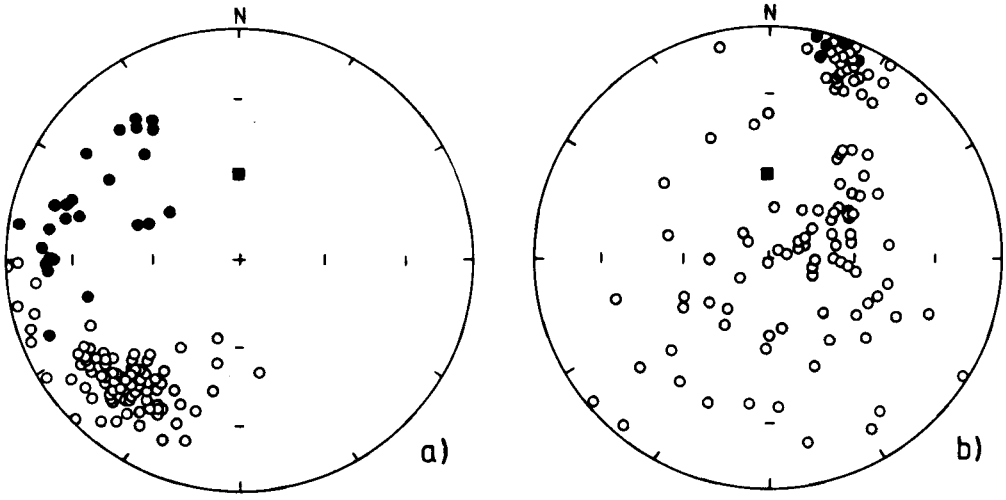


Figure 2. Palaeomagnetic directions observed in the Scisti Silicei Formation before demagnetization. (a) Pignola. (b) Lagonegro. The projection is equal area and the directions are corrected to bedding – the primitive corresponding to the bedding plane. Open symbols are upwards (negative) inclinations. The solid square is the axial geocentric dipole field direction in Lucania.

1365246xb, 1985, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1985.tb04334.x by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

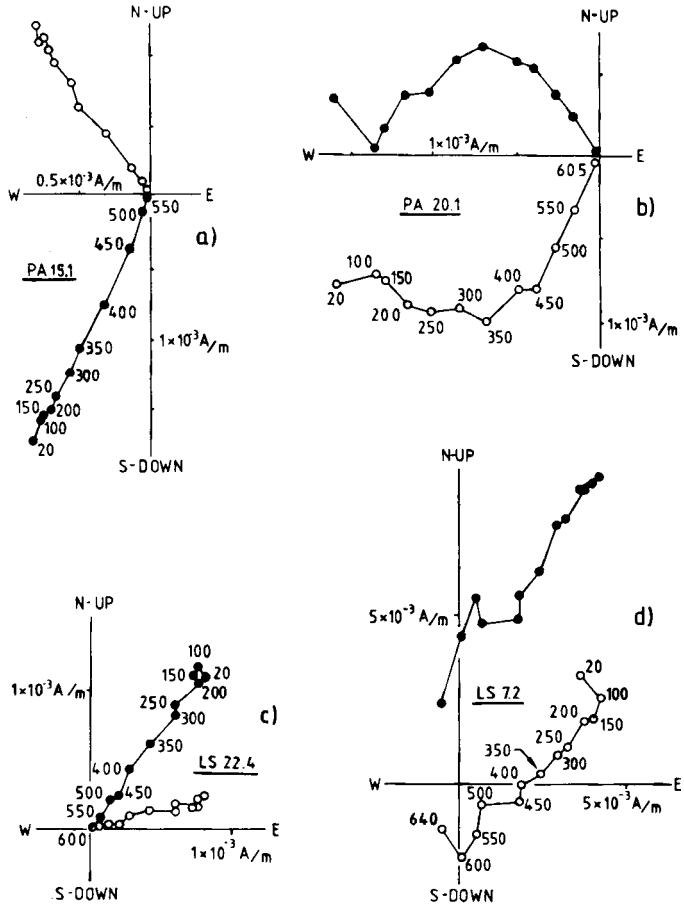


Figure 3. Examples of Cartesian projections of pilot specimen behaviour. Pignola (a, b) and Lagonegro (c, d) pilot specimens show both single component (a, c) and multicomponent (b, d) behaviour in response to thermal demagnetization. Solid symbols, horizontal projection; open symbols, vertical projection. Demagnetization temperatures indicated (°C).

natural remanent intensities between 0.08 and 8.14 mA m^{-1} and directions which formed two main groups; shallow negative inclinations towards the SW and shallow positive inclinations spread between the west and NW (Fig. 2a). The measurable Lagonegro samples had natural remanent intensities between 0.12 and 20.24 mA m^{-1} and generally scattered directions, with only indications of possible groupings (Fig. 2b). At least one sample per site was subjected to incremental thermal demagnetization, with the remanence measured at least twice at each step. The low field susceptibility was also monitored to detect any changes in the magnetic mineralogy. Both single and multiple component behaviour was observed for both localities (Fig. 3) and the high temperature components were not always clearly defined, particularly above 640°C, where rapidly acquired viscous components tended to become dominant over any remaining remanent component. The component analyses for individual specimens were carried out both subjectively and objectively, using the 10° linearity tests suggested by Kirschvink (1980) (Table 1), i.e. the vector was confined within an oblong with a diagonal of 10° or less. The high temperature components were defined by linear behaviour of pilot specimens (Fig. 4), combined with bulk demagnetization of other specimens from the same site at temperatures within the optimum range indicated

1365246x, 1985, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1985.tb04334.x by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

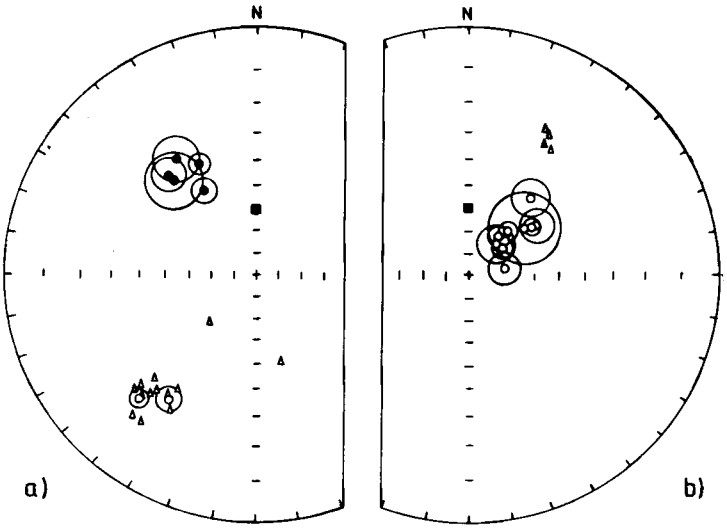


Figure 4. Site mean directions after demagnetization and application of bedding correction. (a) Pignola. (b) Lagonegro. The mean directions combine vectors derived from linear analyses and bulk demagnetized values (see text). Circles represent 95 per cent confidence limits on site mean directions. Symbols as in Fig. 2.

by the pilot specimens. The lower temperature components were defined by linear analyses of the pilot specimen behaviour. Subsequent analyses and discussions have then been confined to site mean vectors, at high or low temperatures, defined with an $\alpha_{95} \geqslant 15°$.

Isothermal remanent properties were used to evaluate the nature of the magnetic mineralogy (Fig. 5). The Pignola specimens characterized by single component remanence showed a sharp increase in magnetization in fields of 0.1 T, tended to saturate by 1.1 T and

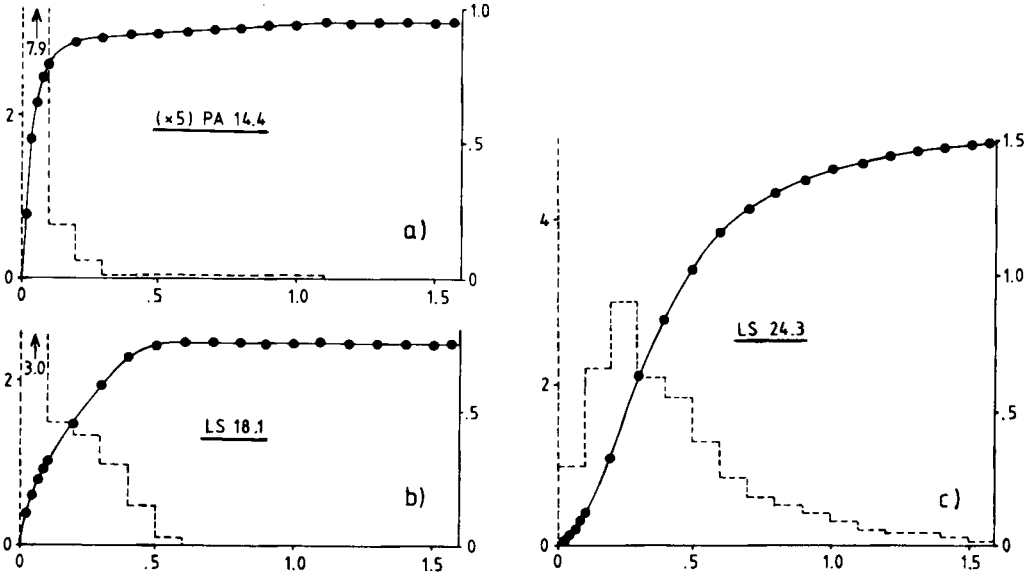


Figure 5. Isothermal remanence acquisition in single component samples. The examples are for (a) Pignola and (b, c) Lagonegro. The horizontal axes are applied field (T) and the vertical axes are intensity (solid symbols, A m^{-1}) and rates of acquistion (dashed lines, A m^{-1} T).

1365246x, 1985, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1985.tb04334.x by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Mean site directions and linear vectors. The left column lists all site mean values ($N = 6$), after demagnetization at the temperatures stated, which have an α_{95} less than 15°. The right column lists all vectors isolated in pilot specimens which show linearity (da = diagonal angle) defined within 10° (Kirkschvink 1980).

Site	°C	Dec.	Inc.	α_{95}	Spec.	°C	Dec.	Inc.	da
PIGNOLA (PA)									
5	500	316.6	32.9	5.7	1.1	100–500	213.0	−32.0	5.4
8	350	316.7	34.6	8.5	3.2	100–400	215.1	−29.0	3.8
14	200	223.3	−22.3	3.1	4.3	200–500	220.1	−28.2	2.9
15	250	214.1	−25.6	4.0	5.1	150–400	217.7	−16.5	5.5
16	500	322.8	30.4	7.1	9.5	150–400	222.2	−26.3	3.0
20	500	324.2	44.7	4.3	10.4	20–350	223.9	−62.8	2.7
22	500	329.4	35.5	3.1	10.4	300–500	223.7	−28.9	6.5
					13.6	200–500	226.3	−25.4	4.5
					16.3	200–300	167.7	−50.3	2.2
					17.4	250–500	226.2	−22.7	6.2
					20.1	100–200	211.9	−25.4	3.6
					23.3	100–400	220.9	−17.1	8.0
					24.4	20–450	223.2	−23.4	3.9
LAGONEGRO (LS)									
8	250	49.3	−68.0	2.5	2.4	200–400	27.8	−22.8	6.4
9	150	52.8	−71.1	5.7	4.1	100–400	29.5	−36.1	6.8
12	150	48.8	−68.6	4.5	5.3	100–400	28.6	−23.6	8.3
13	150	47.4	−68.9	2.5	7.2	200–400	29.0	−26.4	5.0
15	200	51.8	−57.1	14.1					
18	200	78.7	−72.5	7.7					
19	200	53.9	−55.3	3.1					
21	150	53.6	−54.6	6.4					
22	250	42.9	−71.5	8.2					
23	150	43.2	−64.1	4.0					
24	150	38.5	−46.5	7.5					

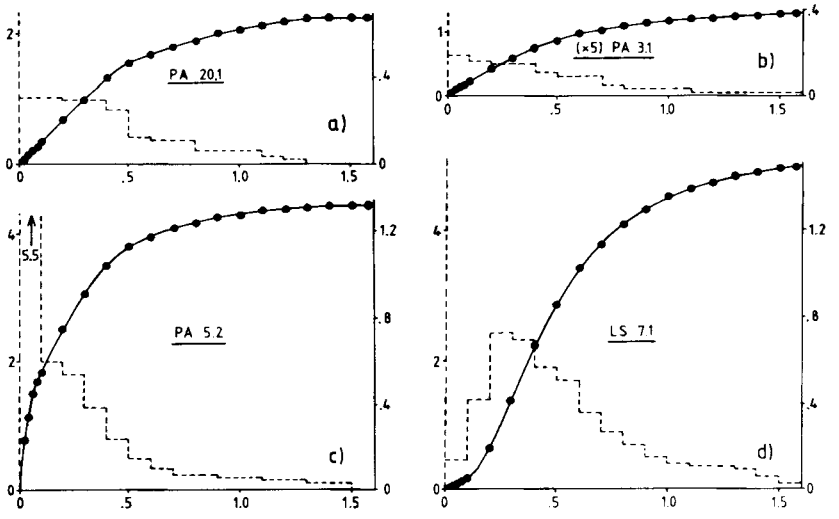


Figure 6. Isothermal remanence acquisition in multicompoent samples. (a, b, c) Pignola. (d) Lagonegro. Axes, etc., as Fig. 5.

1365246x, 1985, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1985.tb04334.x by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

showed coercivity peaks below 0.1 T – all features that suggest that magnetite is the dominant magnetic mineral. In contrast, single component specimens from Lagonegro did not saturate until fields above 0.5 T were applied and some did not saturate at all. The latter were commonly characterized by coercivity peaks below 0.1 T, although sometimes extending to 0.6 T, which would suggest that haematite, in the form of specularite, is present in addition to magnetite (Tarling 1983). The Pignola specimens characterized by multicomponent magnetizations saturated at 1.3–1.5 T (Fig. 6a, c) and some did not saturate (Fig. 6b). A wide range of coercivities were exhibited with values less than 0.1 T being predominant in the specimens which saturated at 1.5 T (Fig. 6c). The Lagonegro specimens characterized by multicomponent magnetizations did not saturate (Fig. 6d) exhibiting an intensity/field pattern similar to some single component Lagonegro specimens (Fig. 5c). This means that both magnetite and haematite, in the form of specularite and as a pigment, are present in varying proportions in both the Lagonegro and Pignola multicomponent specimens.

Discussion and conclusions

The steep inclinations which predominate in the Lagonegro locality, and are of restricted occurrence in the Pignola area, are well defined and of consistent reversed polarity. This suggests that they were acquired in a reversed geomagnetic field prior to folding. However, such an inclination is inconsistent with any possible post-Triassic palaeolatitude for this area (*c.* 25–30°) or with any direction which could have been acquired during a tilt at some specific time during the nappe development and emplacement. Thus these directions, while clearly defined, remain of unknown origin.

The sites from Pignola which show single component magnetization are characterized by a south-westerly declination, intermediate negative inclination, which is carried by magnetite, while the north-westerly direction, of intermediate positive inclination, is characteristic of the high temperature component of the multicomponent sites which contain magnetite, specularite and pigmentary haematite. The identification of the SW component in some of the multicomponent sites, together with the more complex mineralogy, indicates that the NW direction probably represents a later acquired magnetization than the SW directed component.

Single component Lagonegro sites show steep negative inclinations, while north-easterly, medium negative inclinations occur in the multicomponent sites which contain magnetite, specularite and pigmentary haematite. Channell's (1975) results from an E–W anticline at Vietri di Potenza show only two sites with acceptable within-site scatter, of which only one (directed 40°, –25°) can be considered to relate to an expectable palaeolatitude. For the Lucania region (Table 2), the results from three allochthonous areas can be considered to possess inclinations, relative to bedding, which correspond in magnitude and sign with

Table 2. Jurassic mean directions from Lucania, Sicily and Tunisia. The autochthonous results from Sicily (Catalano *et al.* 1981; Nardi 1982) and Tunisia (Nairn *et al.* 1981) show similar inclinations, but radically different declinations to the allochthonous Lucania regions of Italy.

Area	Dec.	Inc.	k	α_{95}	Area	Dec.	Inc.	k	α_{95}
Pignola	219.9	–24.9	162	3.2	Sicily	210.6	–27.0	62	8.6
Lagonegro	28.7	–27.2	174	7.0	Tunisia	175.7	–23.1	115	6.3
Potenza	40.8	–25.5	377	8.5					

1365246xb, 1985, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1985.tb04334.x by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

contemporaneous palaeolatitudes of both allochthonous areas in Sicily (Catalano *et al.* 1981; Nardi 1982) and an autochthonous block in Tunisia (Nairn, Schmitt & Smethwick 1981).

The consistency of the palaeolatitudes, but differences in declinations, observed within nappe structures in Lucania, can be simply explained in terms of the observed differences in orientation of the monoclinical axes in different areas. The rotations relative to Africa, within the horizontal plane, would then correspond to -139° and $+147^\circ$ for the northern and southern extremes respectively (positive = clockwise), and $+44^\circ$ for the central region. The fact that both the upper and lower nappes show similar senses of rotations also supports the concept of a general anticlockwise rotation of the entire Southern Apennine chain (Catalano *et al.* 1976). However, the clockwise rotation indicated for the upper nappe at Pignola would be somewhat inconsistent with this model, although this could be related to highly localized effects. The NW Pignola component (321.8°, 35.7°, $\alpha_{95} = 6.6^\circ$) exhibits an inclination which resembles that of sediments on the African palaeomargin (Channell & Tarling 1975; Catalano *et al.* 1976). The anomalous behaviour of the upper nappe at Pignola, deduced from Jurassic data, is also evident in the, presumed Cretaceous, secondary component.

This study therefore indicates that the emplacement of nappe structures can be examined using palaeomagnetic techniques and suggests that most of their gross deformation must take place at ambient temperatures as there is little evidence for significant magnetic overprinting, as would have been expected if the deformation was accompanied by heating to temperatures of 300°C or more. There is also evidence that some of the motions involved took place by translation, without rotation, such as during the Burdigalian, and that the palaeomagnetic results largely reflect tectonic rotations during the Tortonian–Early Pliocene during the opening of the Tyrrhenian Sea.

Acknowledgments

This study was largely financed by the CNR Ufficio Ralazioni Internazionali grants between 1977 and 1979. We are particularly grateful to Professor Bruno D'Argenio for his encouragement and assistance.

Publication No. 22, Department of Sci. Terra, Naples.

References

Argand, E., 1924. La Tectonique de l'Asie, *Proc. int. Geol. Cong.*, **XIII**, 171–372.

Bernoulli, D., 1972. North Atlantic and Mediterranean Mesozoic facies: a comparison, in *International Report on the Deep Sea Drilling Project*, **11**, 801–871, eds Hollister, C. D. *et al.*, US Government Printing Office, Washington, DC.

Bernoulli, D. & Jenkins, H. C., 1974. Alpine, Mediterranean and Central Atlantic Mesozoic facies in relation to the early evolution of the Tethys, in *Modern and Ancient Geosynclinal Sedimentation*, eds Dott, J. R. & Shaver, R. H., *Spec. Publ. Soc. econ. Paleont. Mineral., Tulsa*, **19**, 129–160.

Catalano, R. & D'Argenio, B., 1978. An assay of palinspastic restoration across western Sicily, *Geol. Romana*, **17**, 145–159.

Catalano, R., Channell, J. E. T., D'Argenio, B. & Napoleone, G., 1976. Mesozoic paleogeography of the Southern Apennines and Sicily. Problems of paleotectonics and paleomagnetism, *Mem. Soc. geol. ital.*, **15**, 96–118.

Catalano, R., D'Argenio, B., Gregor, G. B., Nairn, A. E. M. & Nardi, G., 1981. A preliminary paleomagnetic study of the Ammonitico Rosso of western Sicily, in *Rosso Ammonitico Symposium Proceedings*, pp. 77–89, eds Farinacci, A. & Elmi, S., Tecnoscienza, Rome.

Channell, J. E. T., 1975. Palaeomagnetic studies in Italy and Greece and the evolution of the Mediterranean region, *PhD thesis*, p. 182, University of Newcastle upon Tyne.

1365246xb, 1985, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1985.tb04334.x by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Channell, J. E. T. & Tarling, D. H., 1975. Palaeomagnetism and the rotation of Italy, *Earth planet. Sci. Lett.*, **25**, 177–188.

Channell, J. E. T., D'Argenio, B. & Horvath, F., 1979. Adria, the African promontory in Mesozoic Mediterranean palaeogeography, *Earth Sci. Rev.*, **15**, 213–292.

D'Argenio, B., 1970. Evoluzione geotectonica comparara tra alcune piattaforme carbonatiche dei Mediterranei Europeo ed Americano, *Atti Accad. Pont. Napoli*, **20**, 3–34.

D'Argenio, B., 1976. Le piattaforme carbonatiche Periadriatiche: Una rassegna di problemi nel quadro geodinamico mesozoico dell Area Mediterranea, *Mem. Soc. geol. ital.*, **13**(2), 1–28.

D'Argenio, B., Pescatore, T. & Scandone, P., 1975. Structural pattern of Campania-Lucania Apennines, in *Structural Model of Italy*, eds Ogniben, L. & Praturlon, A., *Quad. Ric. Sci., C.N.R.*, **90**, 313–328.

D'Argenio, B., Horvath, F. & Channell, J. E. T., 1980. Paleotectonic evolution of Adria, the African Promontory, in *Geology of the Alpine Chain, born of the Tethys*, eds Abouin, J., Debelmas, J. & Latreille, M., *Mém. Bur. Rech. Géol. Min. No. 115*, 331–351.

Di Nocera, S., Ortolani, F. & Torre, M., 1975. Fase Messiniana nell'Appennino Meridoniale, *Boll. Soc. Nat. Napoli*. **84**.

Incoronato, A., 1982. Palaeomagnetic studies in the southern Apennines, Italy, *PhD thesis*, p. 182, University of Newcastle upon Tyne.

Lowrie, W., Alvarez, W., Silva, I. P. & Monechi, S., 1980. Lower Cretaceous magnetic stratigraphy in Umbrian pelagic rocks, *Geophys. J. R. astr. Soc.*, **60**, 263–281.

Lowrie, W. & Channell, J. E. T., 1984. Magnetostratigraphy of the Jurassic-Cretaceous boundary in the Maiolica limestone (Umbria, Italy) *Geology*, **12**, 44–47.

Nairn, A. E. M., Schmitt, T. J. & Smethwick, M. E., 1981. A paleomagnetic study of the upper Mesozoic succession in northern Tunisia, *Geophys. J. R. astr. Soc.*, **65**, 1–18.

Nardi, G., 1982. Tectonic and magnetostratigraphic significance of middle and upper Jurassic rocks from western Sicily, *PhD thesis*, p. 78, University of Carolina.

Ortolani, F., 1978. Alcune considerazioni sulle fasi tetloniche Mioceniche e Plioceniche dell'Appennino Meridoniale, *Boll. Soc. geol. ital.*, **97**, 1–8.

Scandone, P., 1972. Studi di geologia Lucana: Carta dei terreni della serie Calcareo-Silico-Marnosa e note illustrative, *Boll. Soc. Nat. Napoli*, **81**, 225–299.

Scandone, P., 1975. Triassic seaways and Jurassic Tethys Ocean in the Central Mediterranean area, *Nature*, **256**, 117–119.

Tarling, D. H., 1983. *Palaeomagnetism*, p. 379, Chapman & Hall, London.

1365246xb, 1985, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1985.tb04334.x by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License